Early Prediction of Work Disability in SLE Patients

Using Machine Learning to Model Labor Force Dropout to Guide Early Clinical Intervention

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Bombardier Research Rounds
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ReSTORE

UHN Schroeder Arthritis Institute

UNIVERSITY OF TORONTO FACULTY OF MEDICINE

Dalla Lana School of Public Health

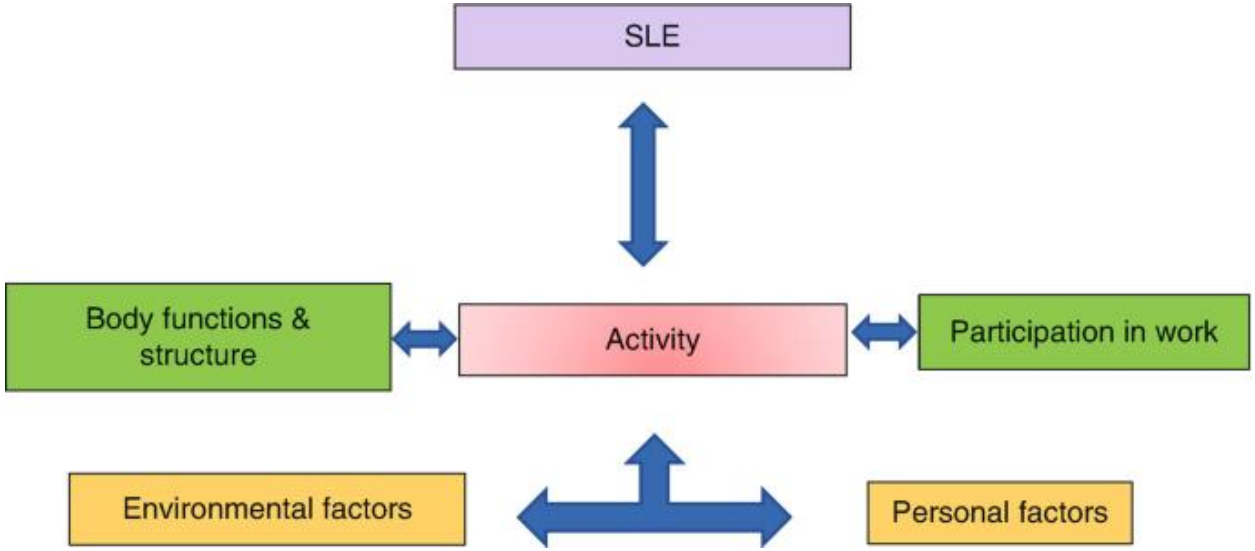
Objectives

1. To create a functional profile of Work Disability (WD) in patients with SLE to inform development of multi-disciplinary evidence-based interventions to prevent WD
2. To review the methodological framework for the development of a risk prediction model
3. To estimate how far in advance of permanent work disability onset a patient's longitudinal clinical data contains a latent signal sufficient to flag those at high risk and understand what factors are driving changes in risk scores

0. SLE and Work Disability

Overview of SLE

- SLE is a chronic autoimmune disease characterized by multisystem involvement
- It manifests itself through mental and physical impairments that disrupt daily life
- The disease course is unpredictable, moving between periods of remission and flares



Work Disability In SLE

- Inability to fulfill job duties
- May lead to early retirement, or prolonged sick leave
- 20–40% of patients exit the workforce prematurely due to SLE

- **Health Impact:**

  Higher pain, fatigue and depressive symptoms and worse cognitive function
- **Psychosocial Impact**:

  Loss of identity, social integration, and purpose
- **Economic Impact**:

  Loss of income, benefits, and long-term career growth

Current Medical Management of SLE

Treatment is Reactive

- Medications are adjusted based on the severity of current symptoms
- If the disease begins to affect vital organs, more aggressive therapies are introduced

Category	Purpose	Common Examples
NSAIDs	Treats pain, fever, and swelling.	Ibuprofen, Naproxen
Antimalarials	It lowers flare risk.	Hydroxychloroquine
Corticosteroids	Rapidly reduces acute inflammation.	Prednisone
Immunosuppressants	Quiets an overactive immune system.	Mycophenolate, Methotrexate, Azathioprine
Biologics	Targeted therapy for complex cases.	Belimumab, Rituximab

What is the daily functional status of SLE Patients?

1. Creating a Functional Profile

Defining the Functional Profile

ADLs

Work functioning

- Individual's capacity to perform job tasks
- Maintain employment
- Manage work demands amid physical or mental health challenges

CIORA Funded Project

Work-Related Disability and Function in Systemic Lupus Erythematosus (SLE): Outcomes of an Exploratory Study from Different Canadian Centers

Behdin Nowrouzi-Kia, Zahi Touma, Antonio Avina-Zubieta, Mary Fox, William Shaw, Maggie Ho, Qixuan Li, Catherine Ivory, Paul Fortin, Stephanie Keeling, Jennifer Reynolds, Derek Haaland, Janet Pope, Lily Lim, Murray Urowitz, Laura Whittall Garcia and Dafna Gladman

The Journal of Rheumatology July 2025, 52 (Suppl 2) 12; DOI: https://doi.org/10.3899/jrheum.2025-0314.POD05

- Using a sample of 404 patients from Canadian centers
- Our project provided an **initial understanding of the disease on a patient's daily functioning**. A primary concern in functional assessment is the evaluation of a patient's ability (function) to engage in work

CIORA ICORA

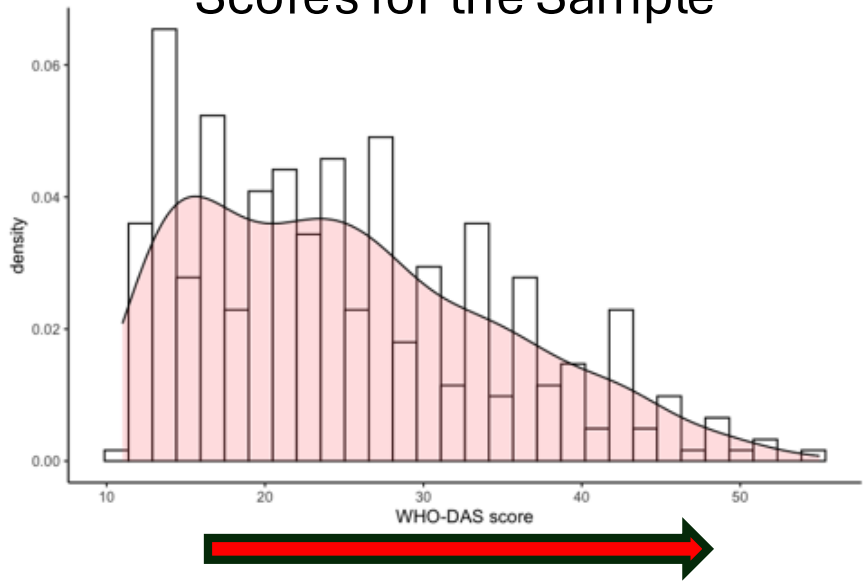
CANADIAN INITIATIVE FOR OUTCOMES IN RHEUMATOLOGY CARE | INITIATIVE CANADIENNE POUR DES RESULTATS EN SOINS RHUMATOLOGIQUES

Measuring Disability with the WHO-DAS 2.0

A 36-item tool assessing six domains: Cognition, Mobility, Self-care, Getting Along, Life Activities, and Participation

- Sample Finidings: Mean score of 25.1
- **This represents a disability level higher than 93.8% of the general Canadian population**
- Significance: High levels of disability exist even when biological disease activity appears low

Histogram of WHO-DAS 2.0 Scores for the Sample



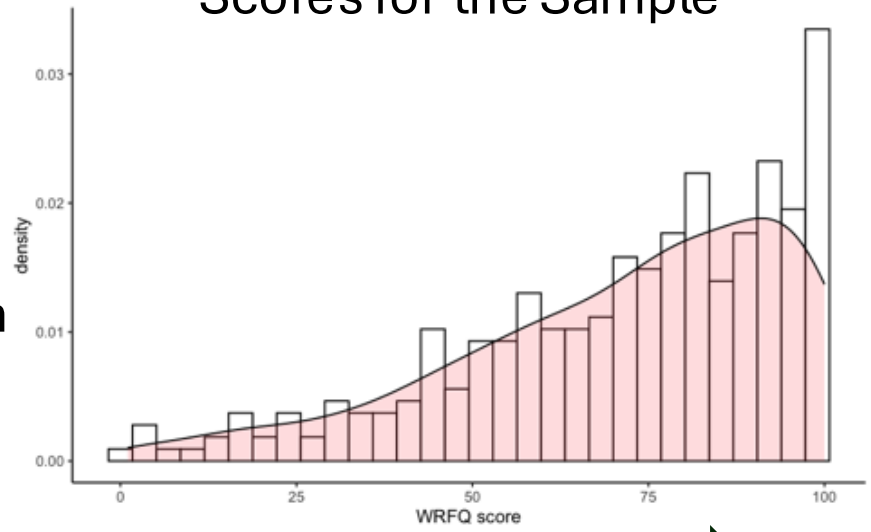
More disability

Measuring Work Productivity with the WRFQ 2.0

A 27-item tool assessing Work Scheduling, Output, Physical, Mental/Social, and Flexibility demands

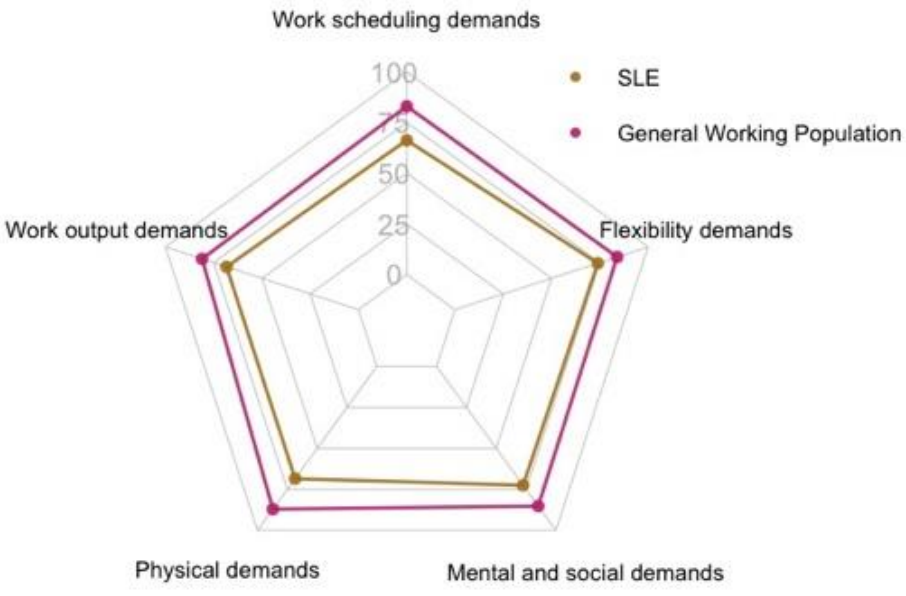
- Sample findings: Mean is 71.5
- Significant deficits found in Physical Demands and Work Scheduling compared to healthy workers
- Significance: SLE functional deficits in physical domains often exceed those found in cancer or common mental disorder populations

Histogram of WRFQ2.0 Scores for the Sample



Better functioning

Mean WRFQ 2.0 Sub-Scale Scores



WRFQ sub-scales	SLE population Mean (SD)	Cancer population Mean (SD)[5]	Common mental disorder population Mean (SD)[5]
Work Scheduling demands	66.8 (28.8)	77.3 (21.3)	65.3 (24.3)
Work output demands	71.1 (25.6)	74.6 (23.0)	64.7 (23.7)
Physical demands	67.3 (27.9)	83.7 (19.3)	90.5 (21.9)
Mental and social demands	74.4 (22.8)	75.4 (21.2)	64.1 (20.5)
Work flexibility demands	75.0 (24.7)	78.4 (20.9)	N/A

5 **The Work Role Functioning Questionnaire v2.0 showed consistent factor structure across six working samples**

Femke I Abma, Ute Bültmann, Benjamin C Amick III, Iris Arends, Heleen F Dorland, Peter A Flach, Jac J L van der Klink, Hardy A van de Ven, Jakob Bue Bjørner

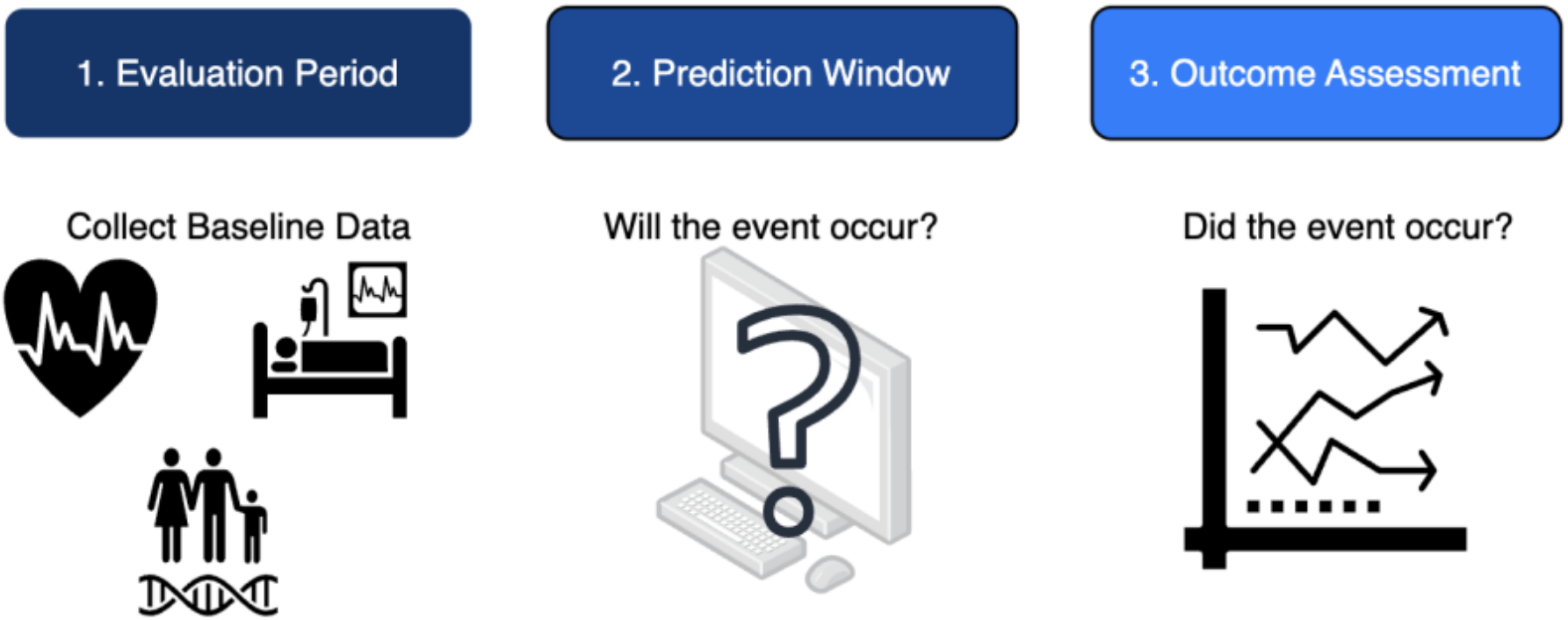
Analysis & Data, The National Research Centre for the Working Environment

Why wait for functional decline?

2. Developing a Risk Prediction Model

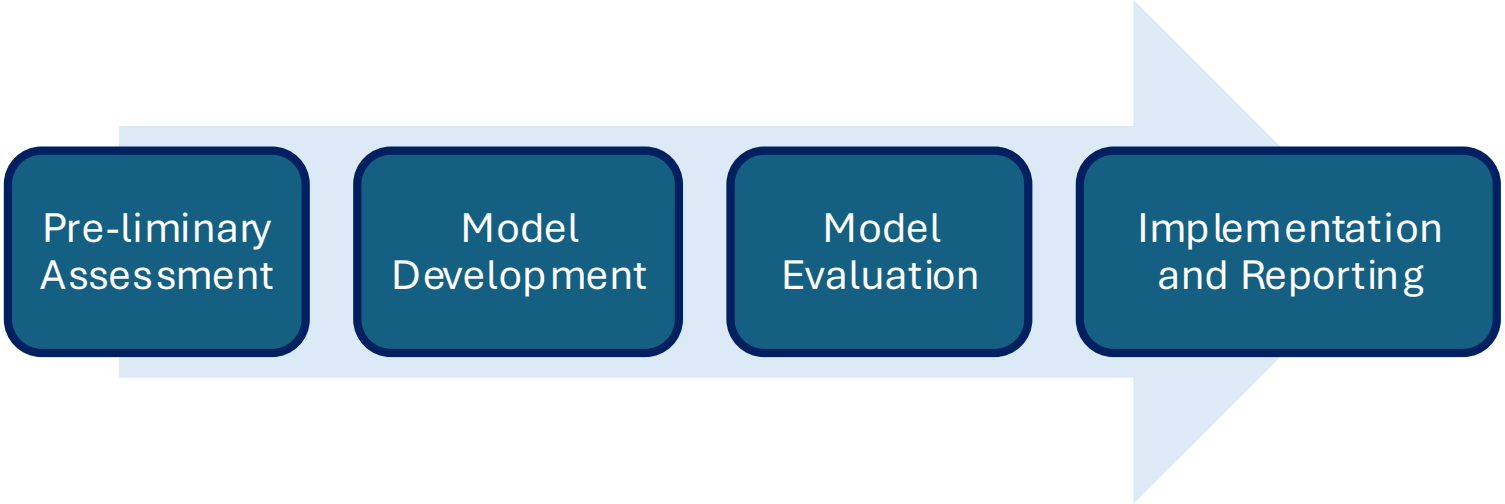
2.0.1 Risk Prediction Models

- Mathematical equations that use patient data to estimate the probability of a healthcare outcome



2.0.2 Stages of Risk Prediction Models

- We will focus on binary outcomes (e.g. Death/Alive or WD/No WD)



Statistical Primer: developing and validating a risk prediction model† FREE

Stuart W Grant, Gary S Collins, Samer A M Nashef

European Journal of Cardio-Thoracic Surgery, Volume 54, Issue 2, August 2018, Pages 203–208, https://doi.org/10.1093/ejcts/ezy180

2.1 Pre-liminary Assessment

- **Evaluating Existing Models**
  Consider whether a new model is needed. Identify, evaluate and potentially update any existing models

- **Candidate Predictors**
  Only consider predictors that have a plausible relationship with the outcome. Use subject (clinical) knowledge and systematic reviews to identify candidate predictors

- **Sample size**
  Events-per-Variable (EPV) guideline: At least ten outcome events of the smallest class for each additional variable
  Example: With n = 2000 if only 150 events, we can have at most 15 predictors (150/10)

2.2.1 Model Development

Model Choice

Ensure it aligns with the nature of the outcome and the intended use case

- Regression
  - Logistic: Binary outcomes
  - Cox Models: Time-to-event
- Machine Learning

Missing data

Need to know why is data missing

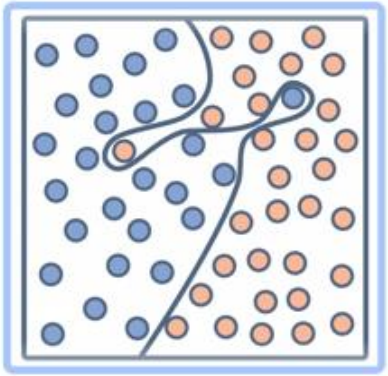
Missing At Random

- The missing value can be predicted by other data
- E.g. a patient accidentally skips a survey question
- Commonly addressed with “multiple imputation”

Missing Not At Random

- The missingness is tied to the value itself
- E.g. high income people refuse to report them

2.2.2 Overfitting



Models often perform excellently on the data used to build them but much worse on new data

- Poorly evaluated models can lead to incorrect clinical decisions and exacerbate healthcare inequalities

Avoid it with:

- Penalization: adds a mathematical penalty to models that try to use too many variables
- Shrinkage: pulls extreme or overly ambitious predictions back toward the average

2.3 Model Evaluation

- Process of assessing how well a prediction model performs
- It determines if the model's predictions are accurate and reliable for a target population

Evaluation of clinical prediction models (part 1): from development to external validation

Gary Collins*, Paula Dhiman, Jie Ma, Michael M Schlussel, Lucinda Archer, Ben Van Calster, Frank E Harrell Jr, Glen P Martin, Karel G M Moons, Maarten van Smeden, Matthew Sperrin, Garrett S Bullock, Richard D Riley
*Corresponding author for this work

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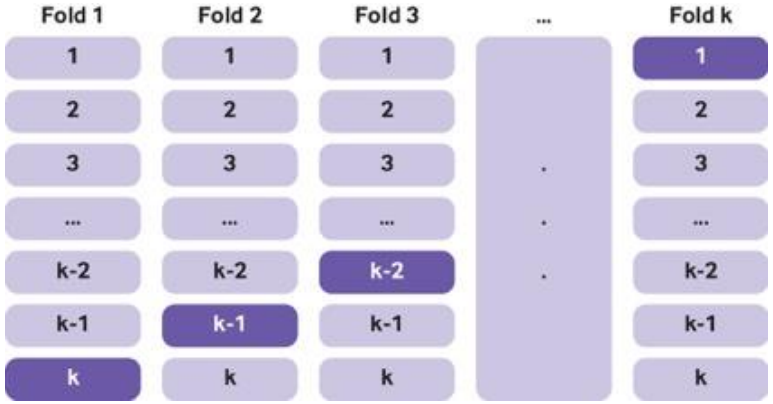
2.3.1 Internal Validation

- Testing the model on the exact same data used for development
- This is typically optimistic and misleading for small samples

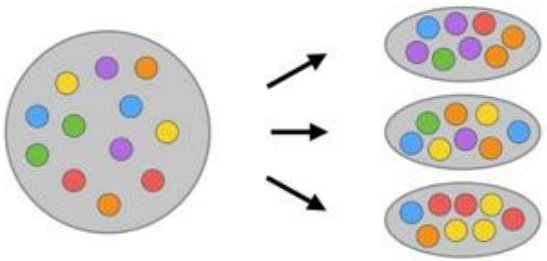
Data Splitting

Resampling Methods

K-Fold Cross Validation

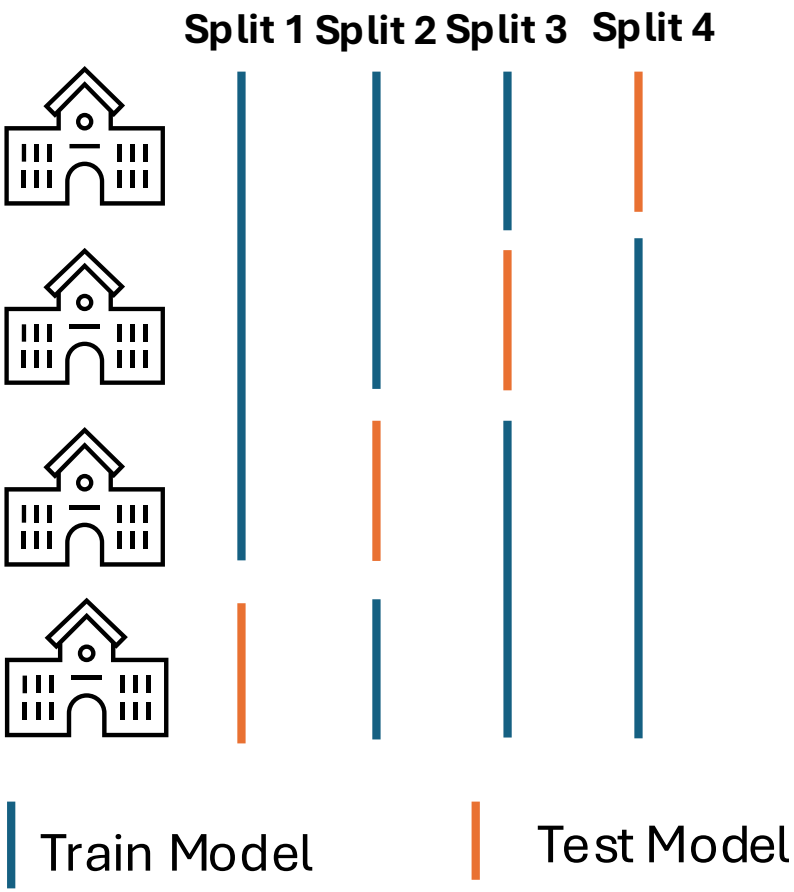


Bootstrapping



An alternative is non-random splitting, which includes splitting by time or location

2.3.2 Internal-External Cross Validation (IECV)



- Uses data from large electronic health record databases, multicenter studies (different clinics), or individual participant data from multiple studies
- Similar to k-fold cross validation but using clusters as the folds

Why use IECV?

- Accounts for heterogeneity across different settings.
- We can show how performance varies by location.

2.3.3 External Validation

Evaluates an existing model using entirely new data that was not available during the model's development

Types of External Validation:

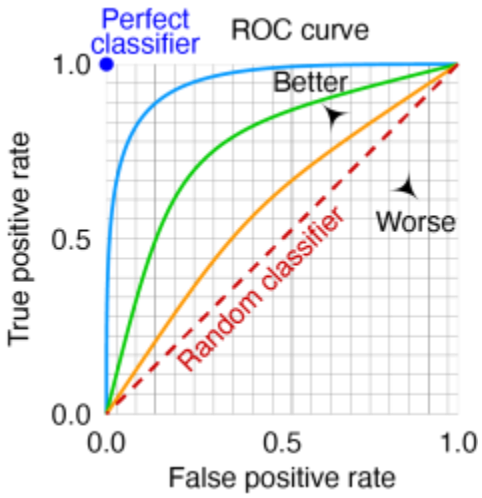
- **Temporal**: Using data from a more recent time period
- **Geographical**: Using data from different centers or countries
- **Transportability**: Testing in different settings (e.g., a model built for adults tested on children)

2.3.4 Measuring Performance

Discrimination

Can the model tell cases apart?

AUC-ROC



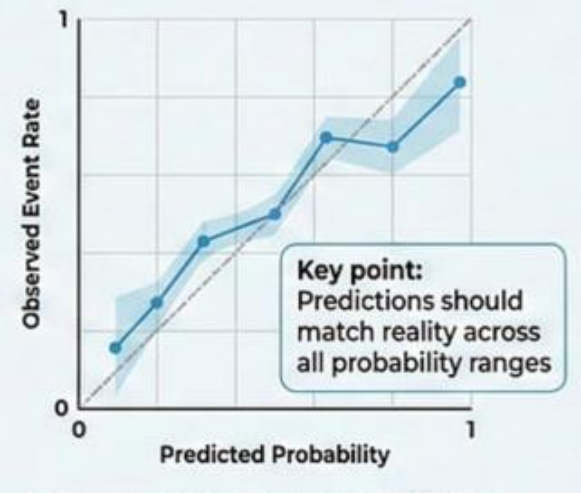
TPR (Sensitivity): proportion of actual positives correctly identified

FPR (Type I Error): proportion of actual negatives that were incorrectly classified as positives

Calibration

Do the predicted risks match reality?

Calibration Plot



Avoid single-threshold metrics like accuracy, which can be misleading

Performance evaluation of predictive AI models to support medical decisions: Overview and guidance

Ben Van Calster, Gary S. Collins, Andrew J. Vickers, Laure Wynants, Kathleen F. Kerr, Lasai Barreñada, Gael Varoquaux, Karandeep Singh, Karel G. M. Moons, Topic Group 6 of the STRATOS Initiative, Tina Hernandez-Boussard, Dirk Timmerman, David J. Mclernon, Maarten Van Smeden, Ewout W. Steyerberg

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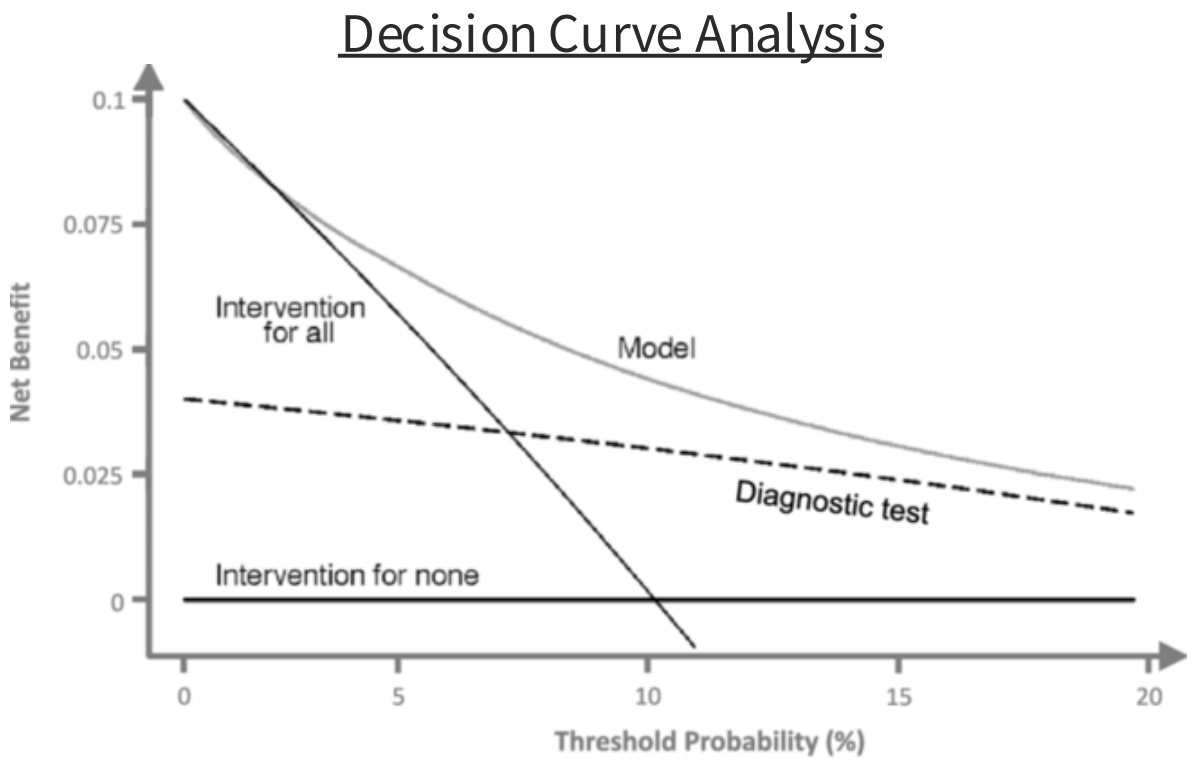
2.4.1 Implementation

- Assess impact through net benefit methods
- Compare model to other models and baseline models flagging all or no patients



2.4.2 Reporting

TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods

Gary S. Collins*, Karel G.M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B. Reitsma, Maarten Van Smeden, Anne Laure Boulesteix, Jennifer Catherine Camaradou, Leo Anthony Celi, Spiros Denaxas, Alastair K. Denniston, Ben Glocker, Robert M. Golub, Hugh Harvey, Georg Heinze, Michael M. Hoffman Show 14 more

*Corresponding author for this work

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TRIPOD+AI

Version: 11-January-2024

Section/Topic	Item	Development / evaluation[1]	Checklist item	Reported on page
TITLE				
Title	1	D;E	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted	
ABSTRACT				
Abstract	2	D;E	See TRIPOD+AI for Abstracts checklist	
INTRODUCTION				
Background	3a	D;E	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models	
	3b	D;E	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (e.g., healthcare professionals, patients, public)	
	3c	D;E	Describe any known health inequalities between sociodemographic groups	
Objectives	4	D;E	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both)	
METHODS				
Data	5a	D;E	Describe the sources of data separately for the development and evaluation datasets (e.g., randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data	
	5b	D;E	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up	
Participants	6a	D;E	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including the number and location of centres	
	6b	D;E	Describe the eligibility criteria for study participants	
	6c	D;E	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant	
Data preparation	7	D;E	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups	
Outcome	8a	D;E	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups	
	8b	D;E	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors	
	8c	D;E	Report any actions to blind assessment of the outcome to be predicted	
Predictors	9a	D	Describe the choice of initial predictors (e.g., literature, previous models, all available predictors) and any pre-selection of predictors before model building	
	9b	D;E	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors)	
	9c	D;E	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors	
Sample size	10	D;E	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation	
Missing data	11	D;E	Describe how missing data were handled. Provide reasons for omitting any data	
Analytical methods	12a	D	Describe how the data were used (e.g., for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements	
	12b	D	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation).	
	12c	D	Specify the type of model, rationale*, all model-building steps, including any hyperparameter tuning, and method for internal validation	
	12d	D;E	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (e.g., hospitals, countries). See TRIPOD-Cluster for additional considerations[1]	
	12e	D;E	Specify all measures and plots used (and their rationale) to evaluate model performance (e.g., discrimination, calibration, clinical utility) and, if relevant, to compare multiple models	
	12f	E	Describe any model updating (e.g., recalibration) arising from the model evaluation, either overall or for particular sociodemographic groups or settings	
	12g	E	For model evaluation, describe how the model predictions were calculated (e.g., formula, code, object, application programming interface)	
Class imbalance	13	D;E	If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions	
Fairness	14	D;E	Describe any approaches that were used to address model fairness and their rationale	
Model output	15	D	Specify the output of the prediction model (e.g., probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified	

How Have We Applied This to SLE?

3. Early Prediction of Work Disability

3.1 Measurable Objectives

1. To estimate the model's accuracy in flagging patients at high risk of work disability

2. To estimate how far in advance of permanent work disability onset a patient's longitudinal clinical data contains a latent signal sufficient to flag those at high risk of workforce exit

3. To understand what factors are the main risk drivers for these patients.

3.2 Defining Workforce Exit

Disability Benefits acts as a Proxy for Functional Decline

- **Actively Employed:** Patients currently working or those reaching standard retirement age while still active in the workforce.
- **Work Disability (WD):** Defined as the receipt of **disability benefits** or prolonged sick leave (> 1 year).
- **Censoring Protocol:** States that need not be directly related to functional decline (e.g., early retirement, economic inactivity, or unemployment), as well as death while employed, are treated as censored.

Robroek, S. J., et al. (2015). Educational Inequalities in Exit from Paid Employment among Dutch Workers: The Influence of Health, Lifestyle and Work
.

3.3.1 Pre-liminary Assessment

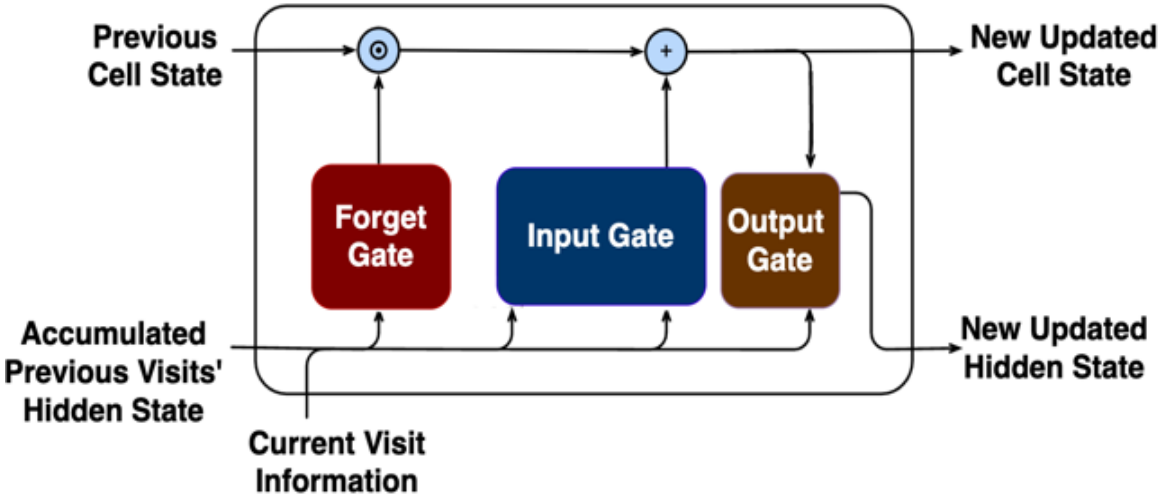
- Current data are records at the Toronto Lupus Clinic (UHN)

  Longitudinal clinical data between 1996-2024

- Variable selection informed with literature review (EPV = 10)

3.3.2 Model Development

LSTM Model Cell For Each Visit



Forget gate filters out unnecessary historical data
Input gate selects relevant information from the new visit
Output gate controls information sent to next cell

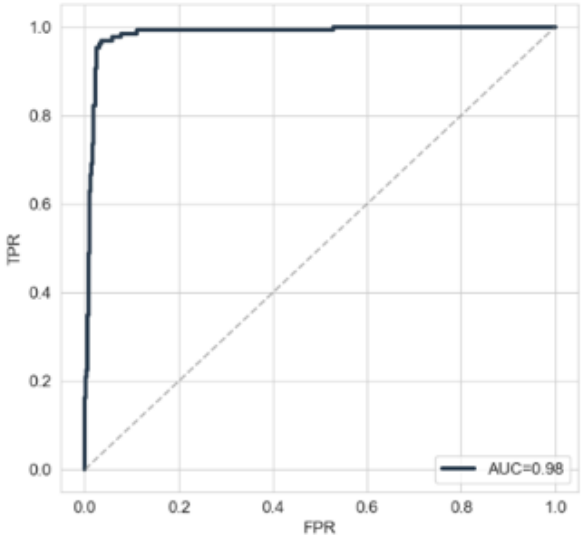
Predictive Risk Modeling Pipeline

- **Internal Validation:** Splitting data into 5-folds for cross validation at the patient level
- **Missing data** imputed using multiple imputation
- Class weighted-loss functions and balanced batching, Ridge regularization

3.4.1 Results: Model Performance

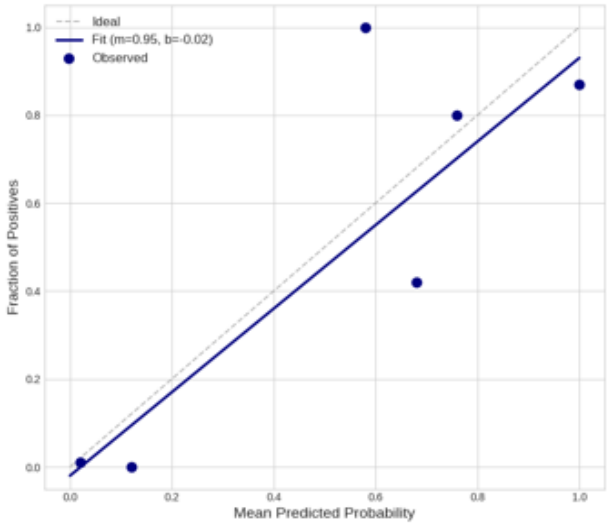
Discrimination

AUC-ROC (0.98)



Sensitivity: 0.89
Specificity: 0.98

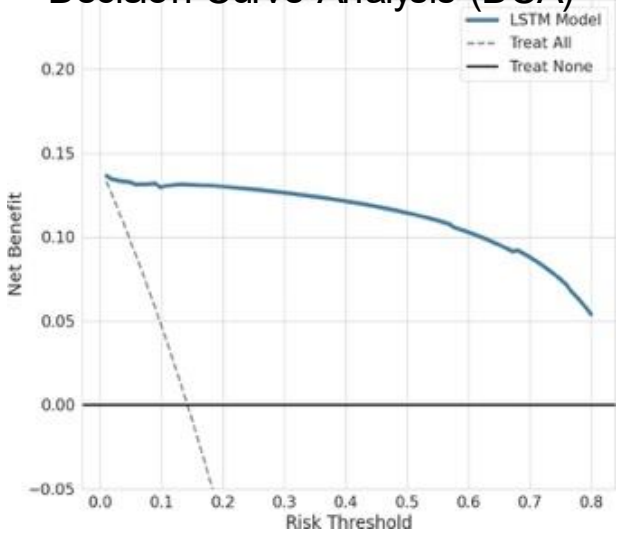
Calibration



Slope for line of best fit = 0.91
Intercept = -0.01

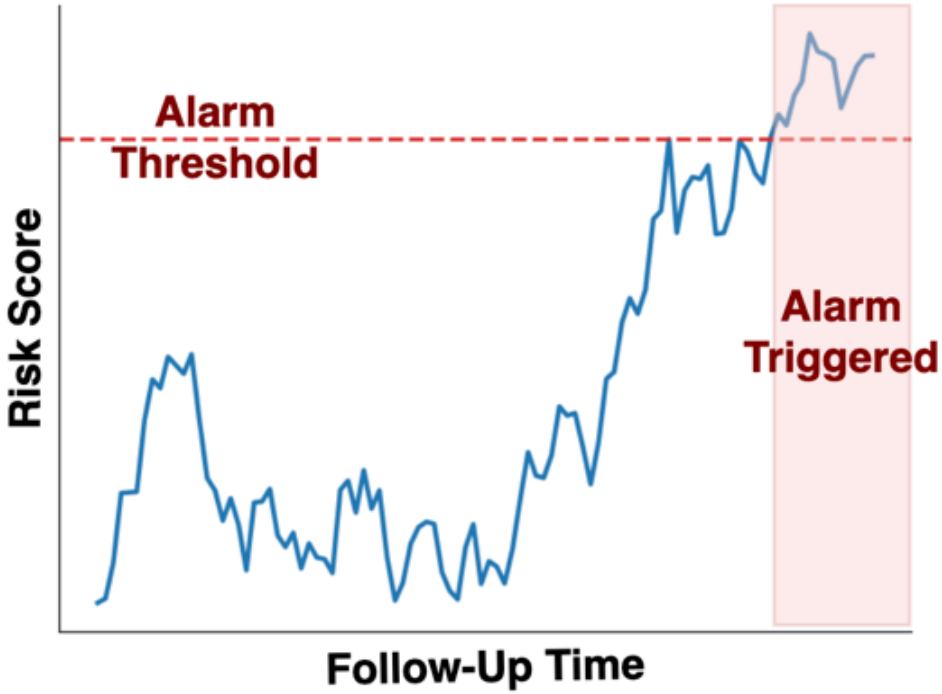
Impact

Decision Curve Analysis (DCA)



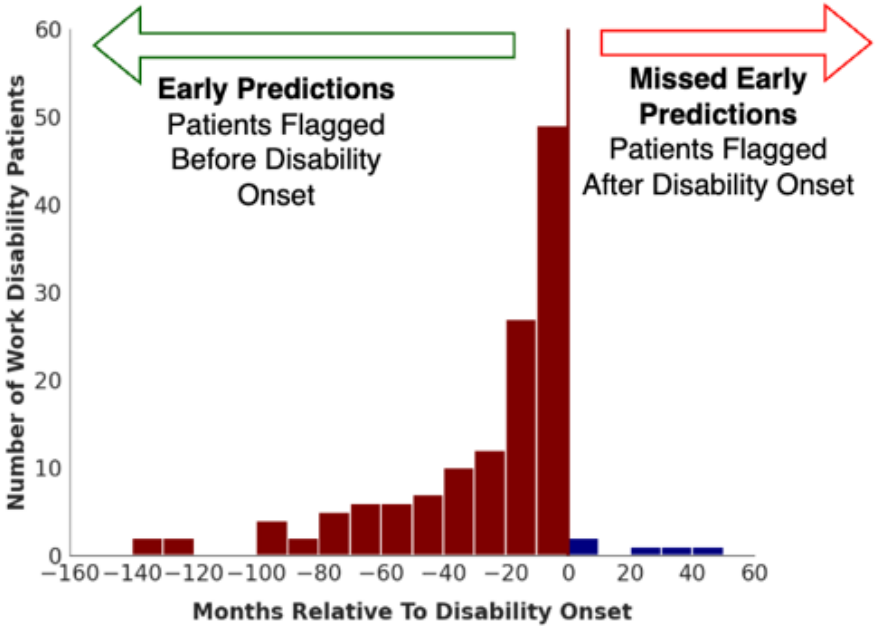
3.4.2 Results: Lead Time & Early Warning

Sample Patient Risk Score Trajectory



Alarm optimized for 89% sensitivity

How early can we predict work disability?

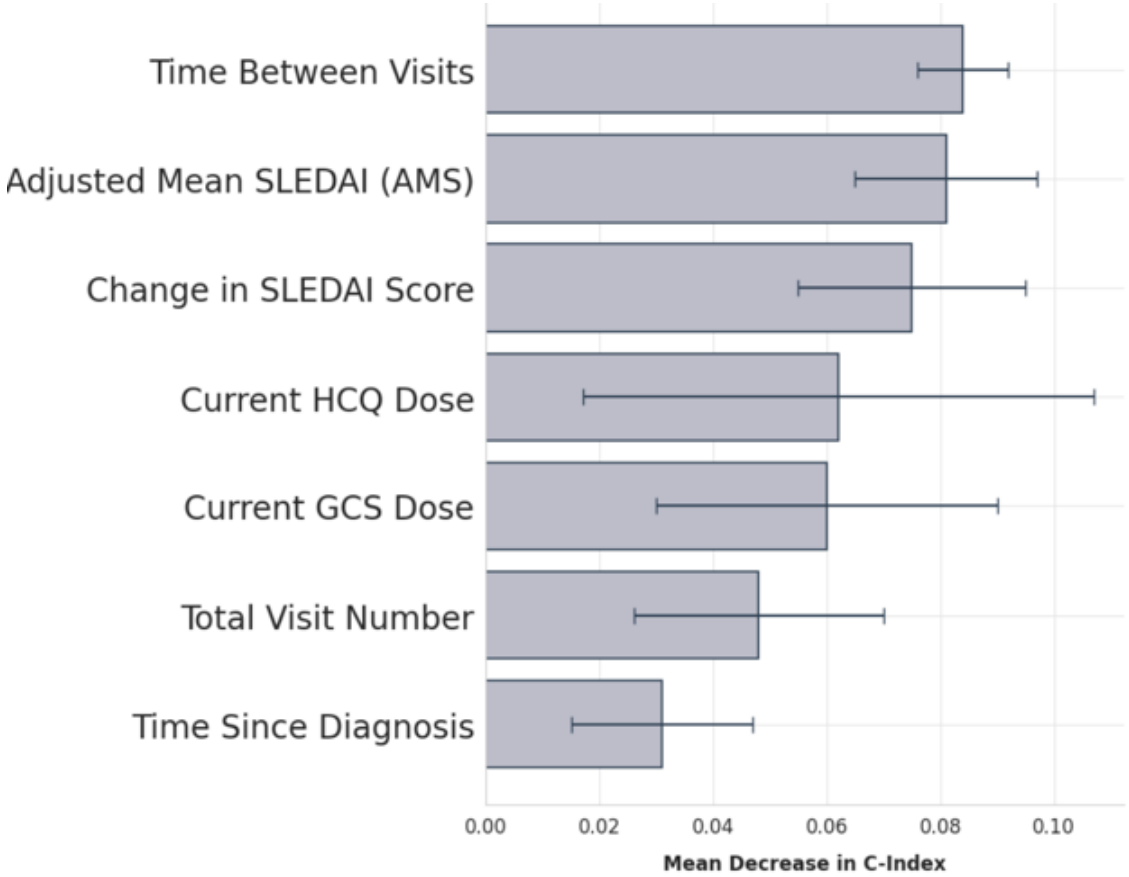


The first alarm was triggered with a median lead time of 28.6 months for correctly predicted future WD cases

3.4.3 Results: Risk Drivers

What variables are driving the risk scores the most?

Permutation-Based Feature Importance Ranking



Plot shows only the seven most influential variables

3.5 Limitations

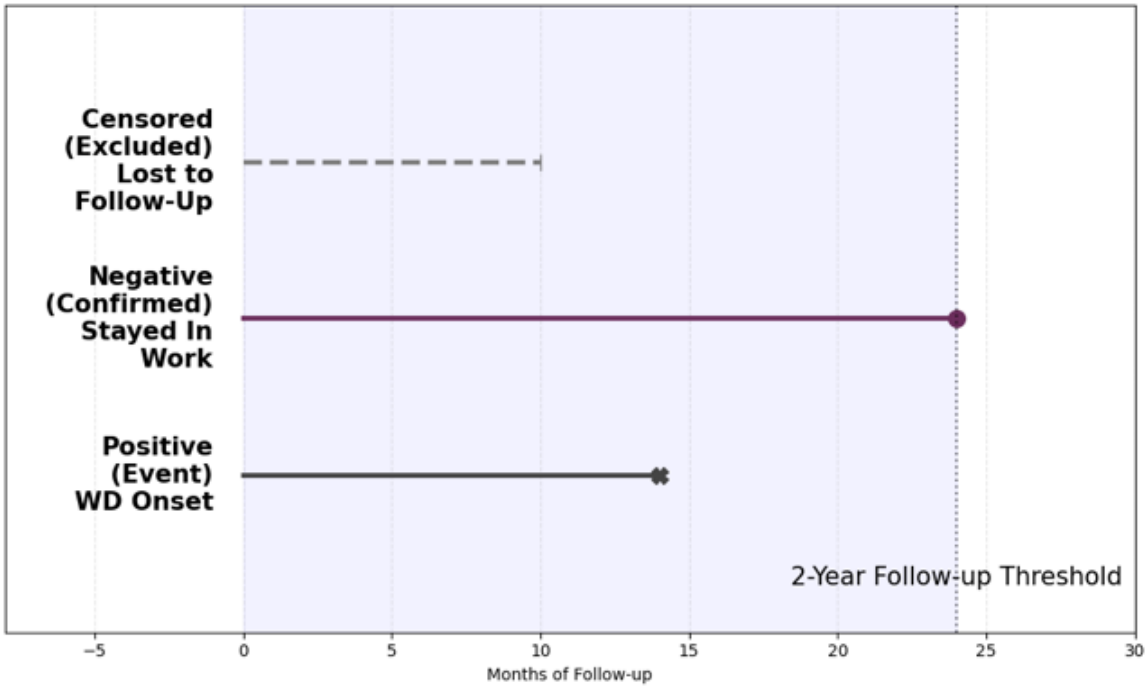
- **Right Censoring Bias**
  We only look at whether the patient reported disability at any time in their follow-up, patients who worked throughout are not right censored

- **Single Site Bias**
  Only using data from the Toronto Lupus Clinic, complicates external validation

Proposed Project

Create a 2-year Work Disability Risk Model

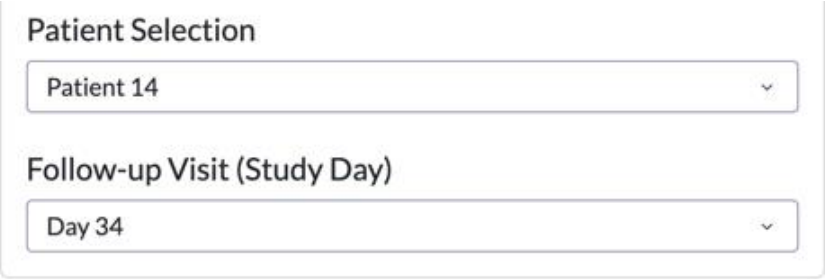
- 2-year window allows us to right censor patients
- Develop and benchmark a set of models using single-center data.
- Compare them by externally validating them with a network of lupus centers
- Enhancing interpretability analysis to see in which direction predictors are driving the risk score in the final model.

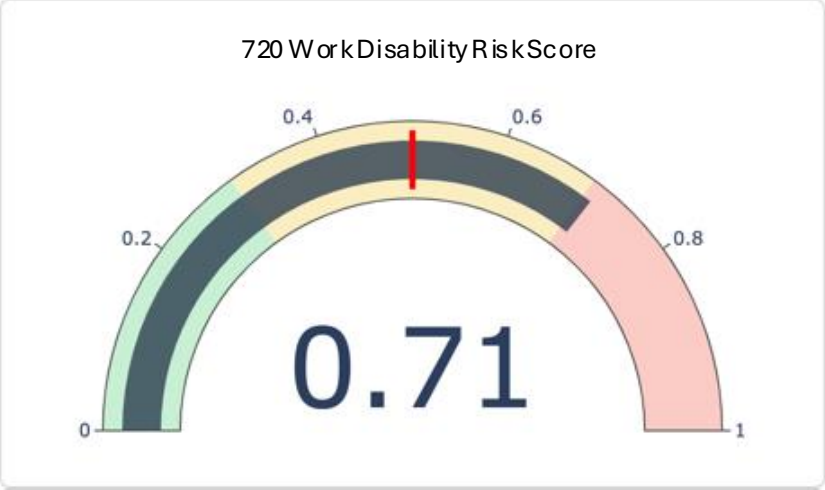


Clinical Relevance: Creating A Tool

Objectives:

- Provide a comprehensive report on a patient's current disease status
- Report historical data to offer summary statistics of other patients' prognoses who were at similar clinical levels
- Enable clinicians to monitor disease progression in real-time and target interventions precisely during the critical window before potential workforce exit





What can be achieved with a 2-year lead time?

Reactive WD Compensation → Change Paradigm → Proactive Workforce Preservation

- Adjusting medications based on the severity of current symptoms.
- Compensating patients unable to work

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Questions?